The Fundamental Theorem of Calculus

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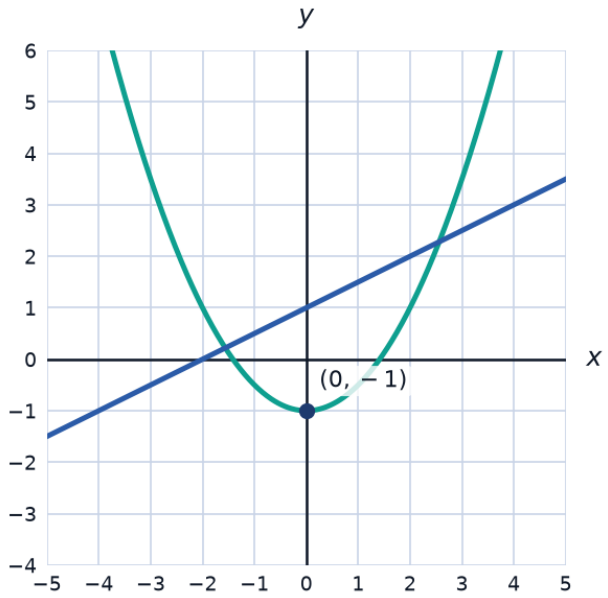
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^3 (x^2 - 2x)\,dx$ using the Fundamental Theorem of Calculus.

2 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_1^4 \sqrt{x}\,dx$.



3 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^{\pi/2} \cos x\,dx$.

4 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^x t^2\,dt$. Find $g'(x)$ using FTC Part 1.

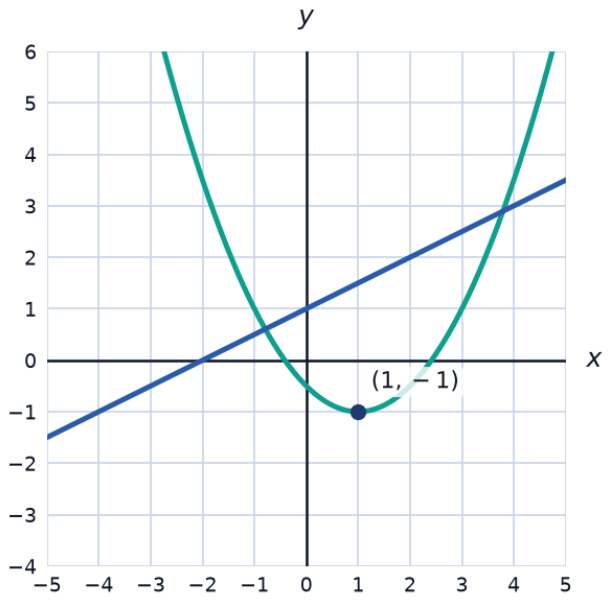
5 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_1^x \sin(t^2)\,dt$. Find $g'(2)$.

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6 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^{x^2} e^{-t}\,dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.



7 [Foundational] • Target: Interpreting accumulation functions

A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau)\,d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Interpreting accumulation functions

Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t)\,dt$, explain what feature of g's graph occurs where f crosses the x-axis, and where f has a local max.

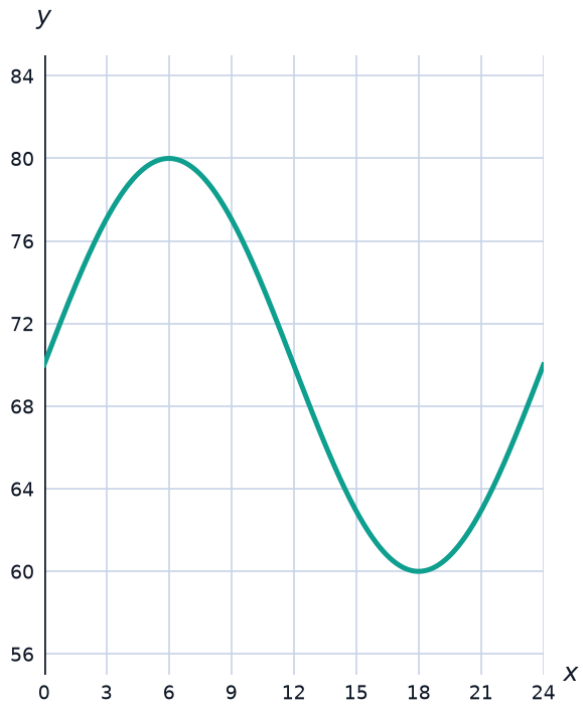
9 [Intermediate] • Target: Average value of a function

Find the average value of $f(x) = x^2$ on $[0, 3]$.

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10 **[Intermediate]** • **Target: Average value of a function**

The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \le t \le 24$. Find the average temperature over this period.



11 **[Intermediate]** • **Target: Definite integrals with substitution-friendly integrands**

Evaluate $\int_0^1 2x(x^2+1)^3\,dx$ using $u = x^2 + 1$.

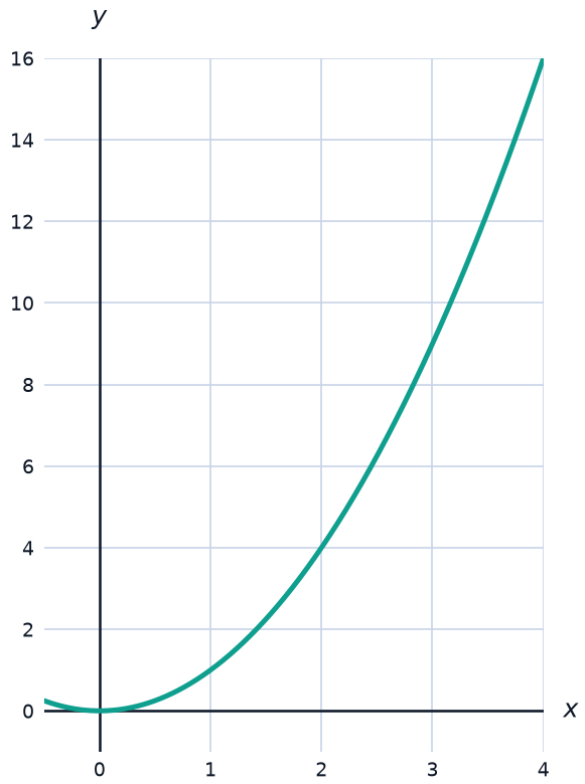
12 **[Intermediate]** • **Target: The Mean Value Theorem for integrals**

Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

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13 **[Intermediate]** • **Target: Visualizing FTC Part 1 as an accumulation function**

The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2\,dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x.



14 **[Intermediate]** • **Target: A capstone FTC application**

Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \le t \le 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

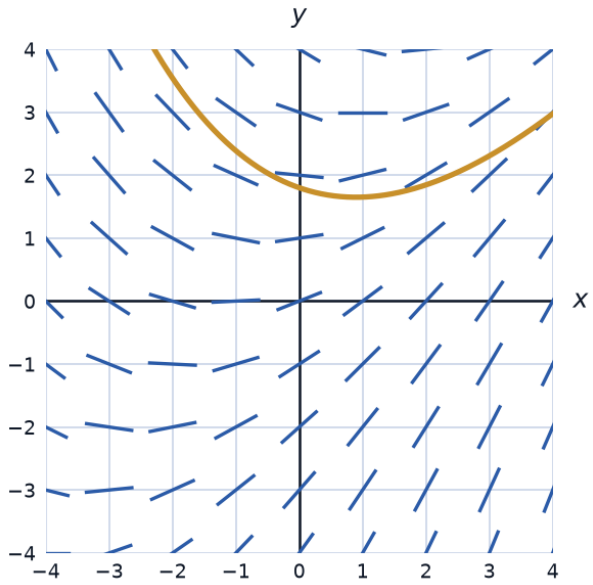
15 **[Challenging]** • **Target: FTC Part 1 with more complex bounds** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Let $g(x) = \int_3^{2x} \ln(t+1)\,dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

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16 [Challenging] • Target: A second average-value application • Cross-Lesson: 8.01 Area Between Curves

A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \le t \le 8$. Find the average production rate over the shift.



17 [Challenging] • Target: Interpreting accumulation functions • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Let $g(x) = \int_{2}^{-3x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

18 [Challenging] • Target: Average value of a function • Cross-Lesson: 1.01 Limits Graphically and Numerically

Let $g(x) = \int_{0}^{-5x} t^2\,dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

19 [Challenging] • Target: Definite integrals with substitution-friendly integrands • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Let $g(x) = \int_{-3}^{4x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

20 [Challenging] • Target: The Mean Value Theorem for integrals • Cross-Lesson: 2.01 The Definition of the Derivative

Let $g(x) = \int_{-2}^{2x} t^2\,dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_{2}^{6} (6x^2 + 8x)dx$ using the Fundamental Theorem of Calculus.

a. 544

b. 546

c. 576

d. 542

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22 [Foundational] • Target: FTC Part 1: accumulation functions

Evaluate $\int_{-1}^{0}(3x^2+2x)dx$ using the Fundamental Theorem of Calculus.

a. -2

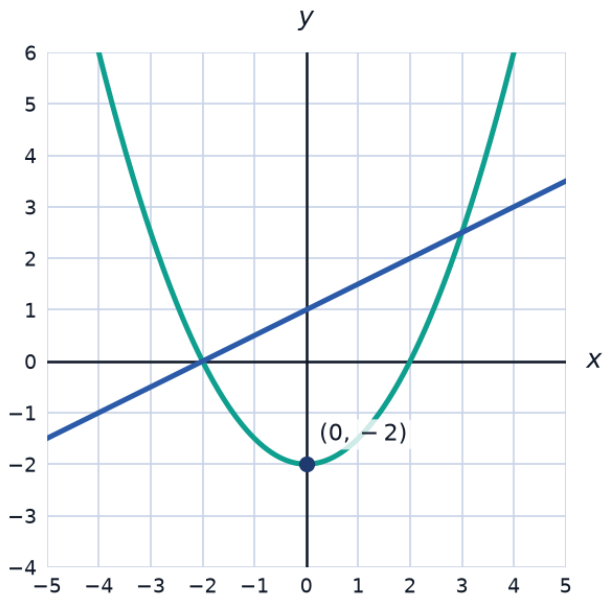
b. 2

c. 3

d. 0

23 [Foundational] • Target: Interpreting accumulation functions

Evaluate $\int_{0}^{2}(-6x^2-2x)dx$ using the Fundamental Theorem of Calculus.



a. -18

b. -22

c. -17

d. -20

24 [Foundational] • Target: Average value of a function

Evaluate $\int_{-2}^{1}(3x^2+2x)dx$ using the Fundamental Theorem of Calculus.

a. 4

b. 8

c. 2

d. 6

25 [Foundational] • Target: Definite integrals with substitution-friendly integrands

Evaluate $\int_{2}^{7}(6x^2+2x)dx$ using the Fundamental Theorem of Calculus.

a. 715

b. 735

c. 717

d. 713

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26 [Foundational] • Target: The Mean Value Theorem for integrals

Evaluate $\int_{-2}^{3}(6x^2)dx$ using the Fundamental Theorem of Calculus.

a. 70

b. 72

c. 54

d. 68

27 [Foundational] • Target: Visualizing FTC Part 1 as an accumulation function

Evaluate $\int_{0}^{5}(6x^2+4x)dx$ using the Fundamental Theorem of Calculus.

a. 298

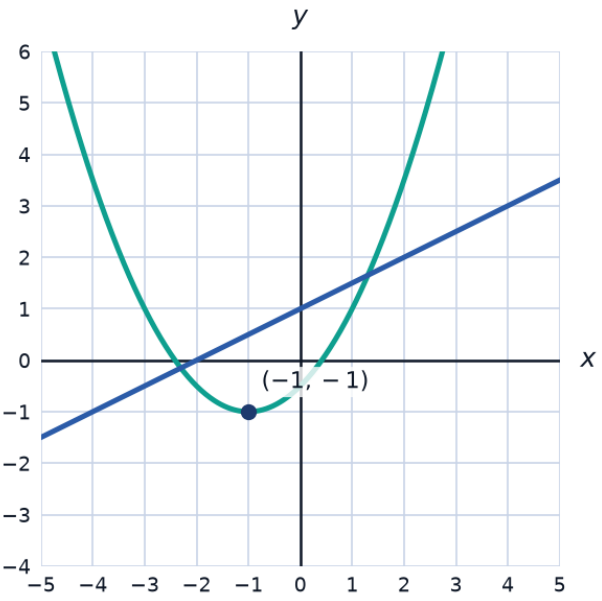
b. 303

c. 302

d. 300

28 [Foundational] • Target: A capstone FTC application

Evaluate $\int_{2}^{5}(3x^2-6x)dx$ using the Fundamental Theorem of Calculus.



a. 50

b. 54

c. 56

d. 52

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: FTC Part 1 with more complex bounds

Let $g(x) = \int_{-3}^{-4x} t^2\,dt$. Find $g'(2)$ using FTC Part 1 and the Chain Rule.

a. -16

b. -256

c. 64

d. -260

30 [Intermediate] • Target: A second average-value application

Let $g(x) = \int_{0}^{3x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. 48

b. 435

c. 432

d. 144

31 [Intermediate] • Target: FTC Part 1 (Differentiating Integrals)

Let $g(x) = \int_{2}^{-4x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. -1028

b. 256

c. -1024

d. -64

32 [Intermediate] • Target: FTC Part 2 (Evaluation Theorem)

Let $g(x) = \int_{3}^{2x} t^2\,dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. 74

b. 36

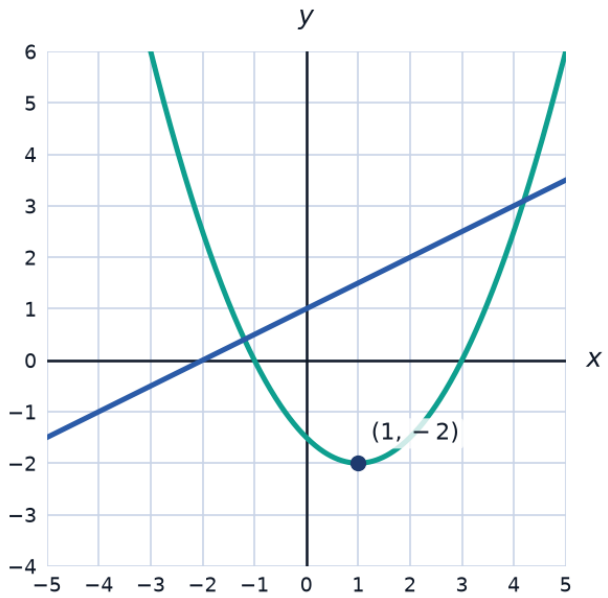
c. 18

d. 72

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33 [Intermediate] • Target: Integration by u-Substitution

Let $g(x) = \int_1^{-5x} t^2\,dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.



a. -2000

b. 400

c. -80

d. -2005

34 [Intermediate] • Target: Average Value of a Function

Evaluate $\int_0^1 2x(x^2+1)^2\,dx$ using $u = x^2 + 1$.

a. 1

b. $\frac{7}{3}$

c. $\frac{7}{2}$

d. $-\frac{7}{3}$

35 [Intermediate] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^1 2x(x^2+4)^2\,dx$ using $u = x^2 + 4$.

a. $\frac{61}{2}$

b. 3

c. $\frac{61}{3}$

d. $-\frac{61}{3}$

36 [Intermediate] • Target: FTC Part 1: accumulation functions

Find the average value of $f(x) = x^2 + 4x + 3$ on $[0, 6]$.

a. 27

b. 63

c. 28

d. 26

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37 [Intermediate] • Target: Interpreting accumulation functions

Find the average value of $f(x) = -3x^2 - 4x$ on $[0, 5]$.

a. -95

b. -36

c. -35

d. -34

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Average value of a function • Cross-Lesson: 1.01 Limits Graphically and Numerically

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and negative for $t < -2$ and positive for $t > -2$. What feature does g have at $x = -2$?

a. An inflection point, since g is always increasing when its integrand is positive.

b. Neither -- accumulation functions never have local extrema.

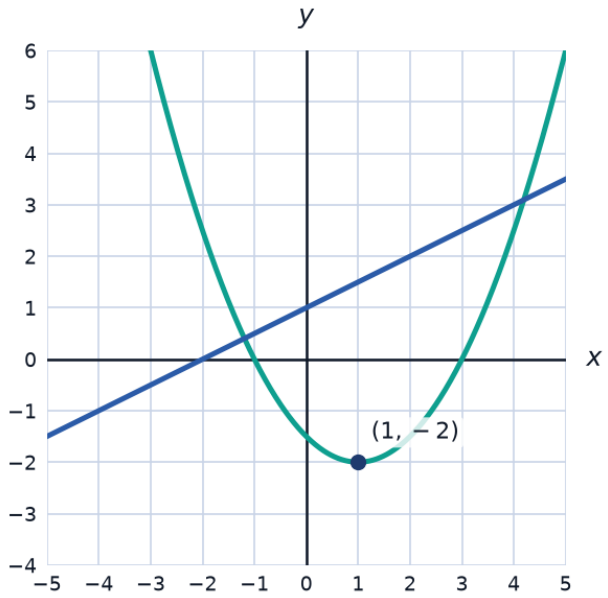
c. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

d. A local maximum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

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39 [Challenging] • Target: Definite integrals with substitution-friendly integrands • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and positive for $t < 1$ and negative for $t > 1$. What feature does g have at $x = 1$?



a. Neither -- accumulation functions never have local extrema.

b. A local minimum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

c. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

d. An inflection point, since g is always increasing when its integrand is positive.

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40 [Challenging] • Target: The Mean Value Theorem for integrals • Cross-Lesson: 2.01 The Definition of the Derivative

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and positive for $t < 5$ and negative for $t > 5$. What feature does g have at $x = 5$?

a. A local maximum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.

b. An inflection point, since g is always increasing when its integrand is positive.

c. A local minimum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.

d. Neither -- accumulation functions never have local extrema.

41 [Challenging] • Target: FTC Part 1 (Differentiating Integrals) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find $\frac{d}{dx}[\int_0^{x^2} \cos(t)dt]$.

a. $2x\cos(x^2)$

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\sin(x^2)$

42 [Challenging] • Target: FTC Part 2 (Evaluation Theorem) • Cross-Lesson: 3.01 The Chain Rule

Evaluate $\int_0^{\pi} \sin x dx$.

a. 2

b. 0

c. 1

d. -2

43 [Challenging] • Target: FTC Part 2 (Evaluation Theorem) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Evaluate $\int_0^1 e^{2x} dx$.

a. $\frac{e^2 - 1}{2}$

b. $e^2 - 1$

c. $\frac{e^2}{2}$

d. $2(e^2 - 1)$

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44 [Challenging] • Target: Integration by u-Substitution • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Evaluate the definite integral $\int_0^1 xe^{x^2}\,dx$.

a. $\frac{e-1}{2}$

b. $e-1$

c. $\frac{e}{2}$

d. $2(e-1)$

45 [Challenging] • Target: Average Value of a Function • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

What is the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 3]$?

a. 6

b. 9

c. 18

d. 3

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